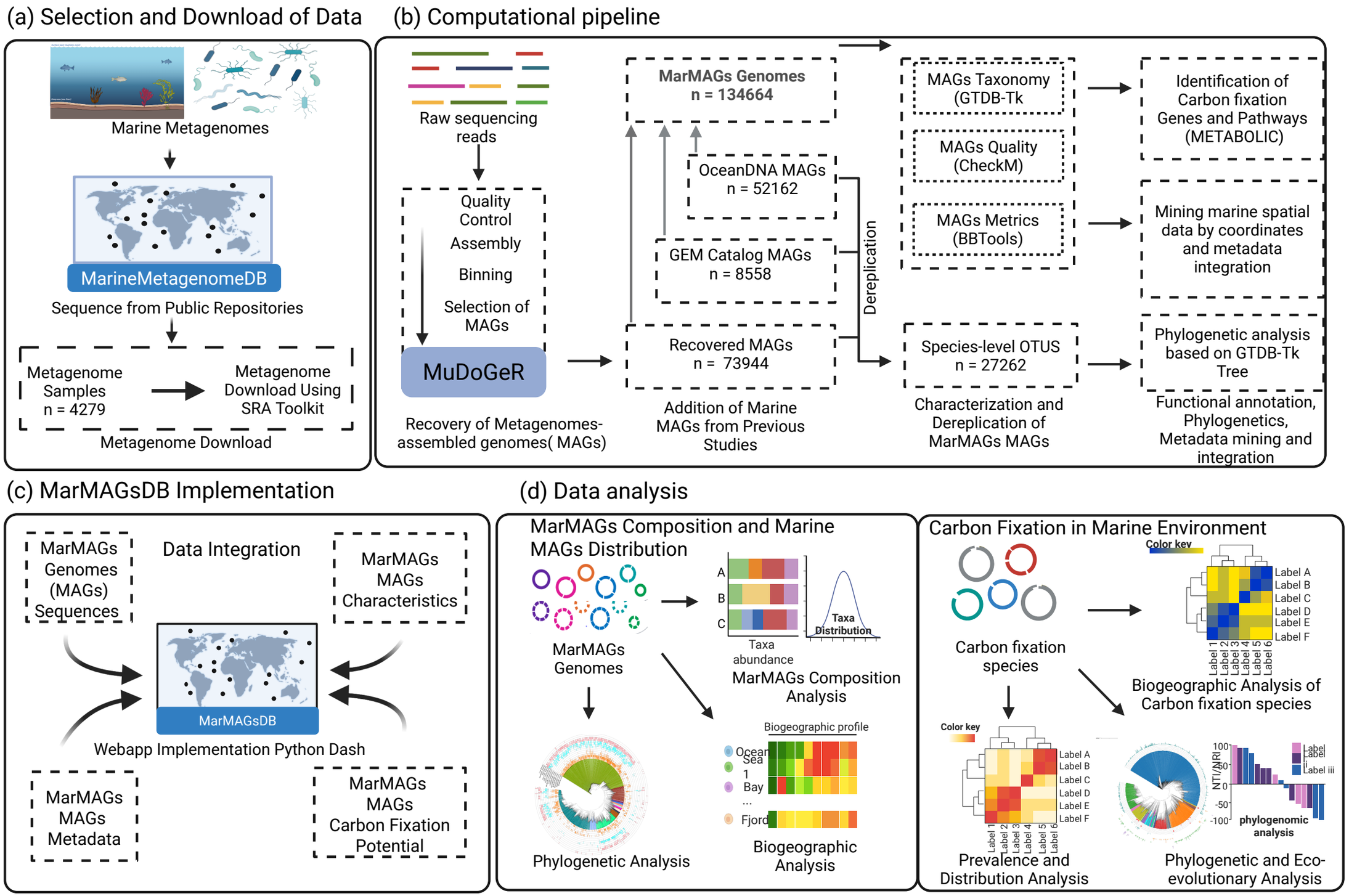
(a) Selection and Download of Data
Marine Metagenomes
MarineMetagenomeDB
Sequence from Public Repositories
Metagenome Samples n = 4279
Metagenome Download Using SRA Toolkit
Metagenome Download
(b) Computational pipeline
Raw sequencing reads
Quality Control
Assembly
Binning
Selection of MAGs
MuDoGeR
Recovery of Metagenomes-assembled genomes(MAGs)
MarMAGs Genomes n = 134664
OceanDNA MAGs n = 52162
GEM Catalog MAGs n = 8558
Recovered MAGs n = 73944
Addition of Marine MAGs from Previous Studies
Dereplication
MAGs Taxonomy (GTDB-Tk
MAGs Quality (CheckM)
MAGs Metrics (BBTools)
Species-level OTUS n = 27262
Characterization and Dereplication of MarMAGs MAGs
Identification of Carbon fixation Genes and Pathways (METABOLIC)
Mining marine spatial data by coordinates and metadata integration
Phylogenetic analysis based on GTDB-Tk Tree
Functional annotation, Phylogenetics, Metadata mining and integration
(c) MarMAGsDB Implementation
MarMAGs Genomes (MAGs) Sequences
Data Integration
MarMAGs MAGs Characteristics
MarMAGsDB
Webapp Implementation Python Dash
MarMAGs MAGs Metadata
MarMAGs MAGs Carbon Fixation Potential
(d) Data analysis
MarMAGs Composition and Marine MAGs Distribution
MarMAGs Genomes
Taxa abundance
Taxa Distribution
MarMAGs Composition Analysis
Phylogenetic Analysis
Biogeographic profile
Ocean
Sea
1
Bay
...
Fjord
Biogeographic Analysis
Carbon Fixation in Marine Environment
Carbon fixation species
Color key
Label A
Label B
Label C
Label D
Label E
Label F
Biogeographic Analysis of Carbon fixation species
Prevalence and Distribution Analysis
NTI/NRI
phylogenomic analysis
Phylogenetic and Eco-evolutionary Analysis